

Getting Started with Scala

1. Installing Scala

For the full details on installing Scala 3, see:

<https://docs.scala-lang.org/scala3/getting-started.html>

What follows below is a summary of what you need.

The simplest approach is to first install the Scala installer, `Coursier`, as described on the link above. Once `Coursier` is installed

`cs setup`

will set up Scala 3 for you. The latest version is 3.7.2. Any version ≥ 3.2 should be fine for this module.

The command

`cs list`

will show what `Coursier` has installed. One important tool you'll be using a lot is **`sbt`** (Scala Build Tool), which is similar to Maven.

2. Setting up a project

In this module we'll use **`sbt`** to create a project. For now, you can create a new project by downloading this simple sbt Scala 3 project:

`https://github.com/UCD-Me1/COMP30950_Greeter`

`cd` to the `COMP30950_Greeter` folder and run the program using:

`sbt run`

Edit `src/main/scala/main.scala` to print something else, and run the program again.

Handy tip: executing **`sbt ~run`** at the command line and every time you save a file in the project, it will be executed again automatically.

3. Using an IDE

You'll want to use an IDE to help with project navigation, code completion etc. The IDE we're using for this module is **IntelliJ IDEA**. The latest version of IntelliJ is **2025.1.4.1**.

To get a licence, browse to <https://www.jetbrains.com/community/education/#students>. An individual license pack will be issued immediately once you sign in with your `@ucdconnect.ie` (or `@ucd.ie`) email address.

It not permitted to use AI tools for programming in this module, so do not install them in your IDE.

Instructions for installing the Scala plugin:

<https://www.jetbrains.com/help/idea/discover-intellij-idea-for-scala.html>

4. Importing an sbt project into IntelliJ

The simplest way to import an sbt project is to use `File > Open` to open the folder containing the `build.sbt` file. This does all the work for you.

To run the program, go to `Run > Edit Configurations...`, click on `Add new` and select `sbt Task`. I suggest writing "sbt run" in the `Name` field, and just write "run" in the `Tasks` field. Leave `Use sbt shell` checked. Now you can run the program by clicking `sbt run` in the IntelliJ title bar.

To run the tests, repeat the process above but using the name "sbt run" and writing "test" in the `Tasks` fields. Make sure the tests run correctly.

Play around with this project, edit the code and the unit tests and run multiple times to get a feeling for the language and the environment.

5. Problems?

Installations may run smoothly, or you may run into problems, possibly due to what you have installed on your laptop already. Some issues to note.

- You should have at least JDK 1.8 (Java 8) installed, but preferably a more recent version. I use version 23.x.
- Ensure that `Settings > Build, Execution, Deployment > Build Tools > sbt` uses the same JRE version as the SDK version in your project (`File > Project Structure > SDKs`)
- If the IDE behaves oddly try deleting the project and target folders in the project. A step further is to do `File > Invalidate Caches...` to reindex the project from scratch.

It's perfectly fine and advised to use AI tools to help with installation problems. The wisdom of the masses embodied in ChatGPT or Claude far exceeds what any demonstrator will know.

Note that the Scala plugin has its own parser to do syntax checking etc., and this may falsely highlight errors (`File > Invalidate Caches...` may fix this). Once your program runs with sbt using Scala 3 as described here, it's fine -- this is how assignments will be tested and graded.

6. Other Tools

Finally two tools for playing around with Scala that are worth a mention:

- * **Scala REPL** (“Read-Evaluate-Print-Loop”): <https://docs.scala-lang.org/scala3/book/taste-repl.html>. The REPL is popular for playing with Scala.
- * **Scatsie**: <https://scastie.scala-lang.org>: The REPL is a command-line interpreter that you can use to test Scala code fragments. Scatsie allows you to run any Scala code in a browser.